

Similar dynamic tibiofemoral movements during jump-landing and walking but distinct hamstrings strength across ACL reconstruction autograft types at mid-term follow-up

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Abstract

Purpose: Dynamic tibiofemoral movements following anterior cruciate ligament reconstruction (ACLR) with quadriceps tendon (QT) autografts have not yet been compared to other autografts. We aimed to compare dynamic anterior tibial translation (ATTd) and internal tibial rotation (ITRd) during jump-landing and walking among patients with QT, bone-patellar tendon-bone (BPTB) and hamstrings tendon (HT) autografts at mid-term follow-up. The secondary aim was to compare lower extremity strength among autografts.

Methods: Patients were included from a previous randomized controlled trial (randomly received primary ACLR with a QT, BPTB or HT autograft), and were prospectively assessed for additional analyses. Primary outcome measures were side-to-side differences in ATTd and ITRd during the side hop, single-leg hop for distance, triple hop for distance, double-leg drop-jump and walking, measured using 3D optoelectronic motion capture, and statistically analyzed using statistical parametric mapping. Moreover, isometric hamstrings and quadriceps strength limb symmetry index (LSI) were assessed using a dynamometer.

Results: Thirty patients were included at 4.7 ± 1.8 years (range: 1.4–7.2) post-ACLR ($n = 10$ per autograft; age: 31.0 ± 4.8 years). No significant between-group differences were observed for side-to-side difference in ATTd and ITRd during all dynamic tasks ($p > 0.05$). QT and BPTB autograft groups demonstrated greater isometric hamstrings strength LSI (QT = 98.4% vs. BPTB = 99.3% vs. HT = 79.0%; $p = 0.02$) and greater hamstrings/quadriceps strength ratio LSI (QT = 102.0% vs. BPTB = 104.6% vs. HT = 79.6%; $p = 0.01$) compared to the HT autograft, but quadriceps strength was similar among autografts.

Abbreviations: ACL, anterior cruciate ligament; ACLR, anterior cruciate ligament reconstruction; ATT, anterior tibia translation; ATTd, dynamic anterior tibial translation; BPTB, bone-patellar tendon-bone; BW, body weight; DLDJ, double-leg drop-jump; GRF, ground reaction force; HT, hamstrings tendon; H/Q, hamstrings/quadriceps; IRB, institutional review board; ITR, internal tibial rotation; ITRd, dynamic internal tibial rotation; LSI, limb symmetry index; QT, quadriceps tendon; RCT, randomized controlled trial; SH, side hop; SLHD, single-leg hop for distance; SPM, statistical parametric mapping; THD, triple hop for distance; UMCG, University Medical Center Groningen.

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Conclusions: Patients who underwent ACLR with QT autografts demonstrated similar side-to-side differences in ATTD and ITRd compared to patients with BPTB and HT autografts 5 years post-ACLR, without demonstrating graft-harvest-related strength deficits, whereas patients with HT autografts showed persistent hamstrings weakness. Graft selection should consider persistent hamstrings strength deficits in patients with HT autografts, while ATTD and ITRd are not autograft-dependent. These findings support QT autografts as a viable option for ACLR.

Level of Evidence: Level II.

KEYWORDS

ACL reconstruction, autograft types, dynamic tibiofemoral movements, knee laxity, knee stability, quadriceps tendon

INTRODUCTION

Anterior cruciate ligament (ACL) reconstruction (ACLR) restores passive knee stability [34], with similar passive stability reported among commonly used ACLR autografts—hamstrings tendon (HT), bone-patellar tendon-bone (BPTB) and the increasingly utilized quadriceps tendon (QT) [11, 32]. However, the ACL-reconstructed knee has been reported to exhibit instability relative to the uninjured contralateral knee during dynamic activities, specifically in terms of dynamic anterior tibial translation (ATTD) and internal tibial rotation (ITRd) [8], potentially contributing to graft failure and elevating the risk of post-traumatic osteoarthritis [51, 55]. Dynamic tibiofemoral movements are, among other mechanisms, controlled by hamstrings and quadriceps activation [41, 42]. Consequently, autograft-specific strength deficits—quadriceps weakness following QT and BPTB harvest; hamstrings weakness following HT harvest—which alter the hamstrings-to-quadriceps (H/Q) strength ratio [17, 23, 31], may contribute to variations in ATTD and ITRd among autograft types used for ACLR. These potential variations have important clinical relevance, as they may inform graft selection and targeted rehabilitation protocols, to potentially reduce the risk of graft failure and post-traumatic osteoarthritis.

Prior cohort studies—focusing solely on BPTB and HT autografts—reported similar dynamic tibiofemoral movements within the first post-operative year [3, 21, 52, 53]. However, these studies lack mid-term data on dynamic tibiofemoral movements, crucial for understanding how autograft-related differences in kinematics, sustained over an extended period, might contribute to the development of osteoarthritis [51]. Furthermore, prior studies did not concurrently quantify hamstrings and quadriceps strength, although graft-harvest-related differences in strength recovery are associated with dynamic tibiofemoral movements and

could help explain potential differences in ATTD and ITRd among autograft types [17, 23, 42]. Finally, prior studies did not use randomized allocation to the autograft type and did not include the QT autograft in the comparison. Although the increasingly popular QT autograft has been proposed as a viable option for ACLR, demonstrating less harvest site pain compared with the BPTB autograft and better functional outcome scores compared with the HT autograft [32], data regarding knee stability in dynamic situations remain absent. To the authors' knowledge, no study has directly compared dynamic tibiofemoral movements among QT, BPTB and HT autografts within a single randomized cohort at mid-term follow-up.

Therefore, the primary aim of this study was to compare side-to-side differences in dynamic tibiofemoral movements among the three most used ACLR autograft types at mid-term follow-up. The secondary aim was to compare isometric hamstrings and quadriceps strength, dynamic knee flexion and valgus angles, peak ground reaction forces (GRFs), and jumping distance between these autograft types. It was hypothesized that patients who underwent ACLR with a QT autograft would exhibit similar side-to-side differences in ATTD and ITRd during jump-landing tasks and level walking compared to patients who underwent ACLR with a BPTB autograft, whereas both QT and BPTB autografts would differ from patients who underwent ACLR with a HT autograft, due to graft-harvest-related muscle strength deficits.

MATERIALS AND METHODS

Design and patients

A secondary analysis of a multicenter randomized controlled trial (RCT) cohort of patients was performed (ClinicalTrials.gov ID: NCT03073083; institutional review board [IRB] number: NL52749.044.16). In the

original RCT, during the study period from March 2017 to March 2022, patients were randomly assigned to undergo ACLR with either a QT ($n = 50$), BPTB ($n = 42$) or HT ($n = 44$) autograft using sealed envelopes. A subgroup of these patients underwent additional assessments as reported in the present study. Eligible participants from the original RCT were recontacted in random order and were invited to participate in the present study until the target sample size was achieved. The present study was conducted at the University Medical Center Groningen (UMCG) in the Netherlands between 2023 and 2025 and was approved by the UMCG's IRB (number: 2022.349). Prior to enrolment, all participants were informed about the study protocol and signed informed consent. The present study is reported in accordance with the STROBE guidelines for cross-sectional analyses nested within a randomized cohort; CONSORT items are referenced for trial provenance [7, 40].

The inclusion and exclusion criteria remained identical to those of the main RCT (ClinicalTrials.gov ID: NCT03073083). Additional inclusion criteria of the present study consisted of patients who: (1) underwent ACLR at least 1 year before testing, (2) were between 18 and 45 years old and (3) understood the Dutch language. Additional exclusion criteria consisted of patients with: (1) chondral lesion at time of surgery ($>2 \text{ cm}^2$ and $>50\%$ depth), (2) previous osteotomy of the tibia or femur, (3) ACLR graft failure and/or contralateral ACLR, and (4) current self-reported lower limb impairments that would result in an inability to perform jump-landing tasks.

Surgical technique and rehabilitation

ACLR with an ipsilateral QT autograft was performed with an all-inside technique using cortical suspensory fixation. The graft was harvested using a cutting blade to provide a 10-mm wide strip, without patellar bone block. The graft was tensioned with the knee at 0° of knee flexion and manual posterior reduction of the tibia relative to the femur. ACLR with an ipsilateral BPTB autograft was performed using the accessory anteromedial portal technique. The graft was fixed in the femoral and tibial tunnel with interference screws at 20° of knee flexion, maximal manual traction on the distal end of the graft, and manual posterior reduction of the tibia relative to the femur. ACLR with an ipsilateral semitendinosus HT autograft was performed with an all-inside technique using cortical suspensory fixation and a quadrupled graft. The graft was tensioned with the knee at 0° of knee flexion and manual posterior reduction of the tibia relative to the femur. All patients underwent rehabilitation by their own physical therapist, following a structured and criteria-based protocol [49].

Outcome measures and dynamic tasks

The primary outcome measure was the side-to-side difference in dynamic tibiofemoral movements during jump-landing tasks and level walking, specifically ATTd and ITRd. Secondary outcome measures included isometric hamstrings strength, isometric quadriceps strength and isometric hamstrings/quadriceps (H/Q) strength ratio. Additionally, jumping distance, peak vertical and posterior GRF, and dynamic knee flexion and valgus angles during the dynamic tasks were included as secondary outcome measures.

Jump-landing tasks consisted of the side hop (SH), the single-leg hop for distance (SLHD), the triple hop for distance (THD), and the double-leg drop-jump (DLDJ). These tasks were derived from return to sports test battery protocols for ACL-injured patients and were selected as they are designed to evaluate jumping performance, produce substantial forces around the knee, and involve standardized movements that resemble those commonly observed in sports activities [14, 44]. Additionally, overground level walking was performed.

Procedure

Dynamic measurement

Reflective markers were placed according to the marker setup for dynamic tibiofemoral movement measurement that is least affected by soft tissue artefacts [26], and calibration data were captured to establish a rigid body marker configuration for the femur and tibia to minimize the effect of soft tissue artefacts (see Appendix SA for further explanation). Three-dimensional (3D) marker positions were recorded at a frequency of 200 Hz using a 16-camera 3D motion capture system (VICON VANTAGE; VICON Motion Systems Ltd). Based on a previous sensitivity analysis, errors in marker position result in an estimated error of 1.5 mm for ATTd of 0.76° for ITRd [26]. GRF was measured using force platforms (AMTI BMS400600HF-OP-2K system, Advanced Mechanical Technology Inc.; $40 \times 60 \text{ cm}$, 2000 Hz).

For the SH, SLHD and THD, participants stood on one leg, jumped as far as possible, and landed on the same leg, maintaining balance for at least 3 s without sliding. The SH was performed sideways toward the standing leg, the SLHD was performed forwards, and the THD was performed with three consecutive forward hops. The starting leg was randomized. For each of these jumping tasks, three practice jumps were performed first. The median distance of these practice jumps served as the starting distance from the centre of the force platform. For the DLDJ, participants dropped from a 40-cm platform positioned 50 cm from two force

platforms, landed with one foot on each platform, and jumped as high as possible before returning to the platforms. After practising, each jumping task was performed three successful times, by landing with the entire foot on the force platform. For level walking, participants walked 10 m at a self-selected pace, with two force platforms positioned midway. Trials were repeated until each foot struck the platform three times. The order of all tasks was randomized, to control for potential training and fatigue effects.

Isometric quadriceps and hamstrings strength measurement

After completing all dynamic tasks, participants performed maximal voluntary isometric quadriceps and hamstrings contractions with both legs on the Biodex System 4 (Biodex Medical Systems Inc.). Participants were seated with their hip and knee joints flexed at an angle of 90°, and performed two trials consisting of a 5-s maximal quadriceps contraction, a 5-s rest, a 5-s hamstrings contraction, and another 5-s rest. Both legs were measured consecutively, following the same leg order as the jump-landing tasks.

Data processing

Force plate data were down-sampled from 2000 to 200 Hz. Raw 3D marker position data were filtered using a zero-phase Gaussian-kernel convolution filter with a nominal low-pass frequency of 10 Hz, smoothing high-frequency noise without introducing phase distortion [5, 20]. Quadratic spline interpolation with zero lag was used to fill gaps in trials that were smaller than 20 frames. For the quantification of kinematic outcomes, a customized MATLAB (v24.1; The MathWorks Inc.) script was used, based on the method of Taylor et al. [45] and Boeth et al. [2]. See Appendix SA for a detailed step-by-step description.

Data were further analyzed using Python v3.10.9 (Python Software Foundation). For the SH, SLHD and THD, data were analyzed between 0.25 s before initial ground contact (IC) and 0.75 s after IC. IC was defined as the time point when the vertical GRF reached at least 5% of the participant's body weight (BW). For the DLDJ and level walking, data were only analyzed during the ground contact phase prior to take-off and during the stance phase, respectively, defined as the periods during which the vertical GRF exceeded 5% of the participant's BW. To ensure consistency in trial lengths, kinematic data during these phases were re-sampled to 200 frames per trial using interpolation.

Mean ATTd, ITRd and dynamic knee flexion and valgus angle temporal waveforms were computed separately for the injured and non-injured leg, using the

three successful trials of each dynamic task. Then, side-to-side differences as temporal waveforms were computed by subtracting the non-injured side from the injured side. For vertical and posterior GRF, the peak force during the landing was determined using the mean temporal waveforms, normalized to BW, and used to calculate side-to-side differences. For hamstrings strength, quadriceps strength and H/Q ratio, the peak torque during the maximal contractions was selected as maximal isometric strength. For isometric strength and jumping distance, the limb symmetry index (LSI) was calculated as $\frac{\text{injured side}}{\text{non - injured side}} \times 100\%$.

Statistical analysis

Sample size calculation

An a priori sample size calculation was performed based on the effect of rectus femoris muscle activity on ATTd ($F=9.12$) [24], as no between-group data for ATTd or ITRd were yet available. With a significance level of 0.05 and 80% power, 10 participants per group were required.

Outcome comparisons

Python v3.10.9 was used for statistical analysis of the outcome measures. For kinematic and kinetic data, only side-to-side differences and LSI values were tested for statistical significance. Between-group comparisons solely for the injured or non-injured leg were not performed.

Discrete outcomes (patient characteristics, isometric hamstrings and quadriceps strength, jumping distance and peak GRF) were assessed for normality using Shapiro–Wilk tests. For normally distributed data, one-way ANOVA was performed with autograft type (QT vs. BPTB vs. HT autograft) as between-subjects factor. When the ANOVA yielded a significant result, post hoc Tukey's HSD tests with p value correction were conducted. For non-normally distributed data, Kruskal–Wallis tests were performed, followed by post hoc Dunn's tests with Benjamini–Hochberg correction in case of a significant main effect. For categorical patient characteristics, chi-squared tests were performed, followed by pairwise chi-squared tests with Benjamini–Hochberg corrections in case of a significant main effect. A p value of <0.05 was considered significant.

For time-series data (ATTd, ITRd, dynamic knee flexion and valgus angle over the time course of the imposed tasks), the open-source `spm1d` package (v0.4; Pataky [35], www.spm1d.org) in Python v3.10.9 was used. First, the normality of the data distribution

was assessed using a statistical parametric mapping (SPM) Shapiro–Wilk test. For normally distributed data, a one-way SPM ANOVA was performed for each outcome measure and each dynamic task. For non-normally distributed data, a statistical non-parametric mapping one-way ANOVA with permutations was conducted [36]. The dependent variables included side-to-side differences in ATTD, ITRd, dynamic knee flexion angle and dynamic knee valgus angle. The between-subjects factor was autograft type (QT vs. BPTB vs. HT autograft). When the SPM test statistic exceeded the critical F value based on an alpha of <0.05 , the null hypothesis was rejected, implying a significant difference between autograft types. When a significant difference was observed, post hoc SPM independent t -tests with Bonferroni corrections were performed.

To quantify effect sizes for ATTD and ITRd, partial η^2 values were computed separately for each task. For each time point, the F -statistic from the one-way SPM ANOVA and the corresponding degrees of freedom were used. Partial η^2 was computed using the following formula [28]:

$$\eta_p^2(t) = \frac{F(t) \times df_{\text{groups}}}{F(t) \times df_{\text{groups}} + df_{\text{total}}}$$

The resulting values were averaged across all time points within the task, yielding a mean partial η^2 for each task. Effects were interpreted as small ($\eta^2 < 0.06$), medium ($\eta^2 \geq 0.06$) or large ($\eta^2 \geq 0.14$) [28].

RESULTS

Of the 136 eligible patients, 28 patients (21%) were excluded due to ACLR graft failure and/or contralateral ACL injury, 2 patients (1%) were excluded due to age, and 29 patients (21%) declined to participate (due to lack of time, travel distance, pregnancy or current injury). Thirty patients were included with a mean age of 31.0 ± 4.8 years (range: 24–42) and a mean follow-up of 4.7 ± 1.8 years (range: 1.4–7.2) after surgery ($n = 10$ per autograft type). Table 1 reports all patient characteristics, with no significant differences among groups.

Primary outcome measure

Dynamic tibiofemoral movements: ATTD and ITRd

Figure 1 displays ATTD in the injured and non-injured leg during all dynamic tasks for each group separately. Figure 2 displays ITRd in the same manner. For both outcome measures and during all dynamic tasks, the QT autograft group demonstrated similar side-to-side

differences compared to the BPTB and HT autograft groups, while the BPTB and HT autograft groups also demonstrated similar side-to-side differences compared to each other ($p > 0.05$). For ATTD, the mean effect sizes across the timeframe of the tasks were small (partial η^2 : SH = 0.05; SLHD = 0.02; THD = 0.05; DLDJ = 0.02; Walking = 0.02). For ITRd, the mean effect sizes across the timeframe of the tasks ranged from small to medium (partial η^2 : SH = 0.05; SLHD = 0.05; THD = 0.06; DLDJ = 0.06; Walking = 0.02).

Secondary outcome measures

Isometric hamstrings and quadriceps strength

Table 2 reports the maximal isometric hamstrings and quadriceps strength and H/Q ratio. For hamstrings strength LSI, post hoc analyses revealed a similar LSI among the QT and BPTB autograft groups, whereas both groups demonstrated greater hamstrings strength LSI compared to the HT autograft group (QT = 98.4%; BPTB = 99.3%; HT = 79.0%; $p = 0.02$). In contrast, all autograft groups demonstrated similar quadriceps strength LSI (QT = 98.8%; BPTB = 95.6%; HT = 98.9%; $p = 0.85$). For H/Q ratio LSI, post hoc analyses revealed a similar LSI among the QT and BPTB autograft groups, whereas both groups demonstrated greater H/Q ratio LSI compared to the HT autograft group (QT = 102.0%; BPTB = 104.6%; HT = 79.6%; $p = 0.01$).

Jumping distance and peak GRF

Table 3 reports the jumping distance and peak GRF. All autograft groups demonstrated similar jumping distance LSI and side-to-side differences in peak posterior GRF ($p > 0.05$). For side-to-side differences in peak vertical GRF, post hoc analyses revealed similar values among the QT and BPTB autograft groups and among the QT and HT autograft groups, whereas the BPTB and HT autograft groups significantly differed (QT = 0.14BW; BPTB = -0.07BW; HT = 0.42BW; $p = 0.04$).

Dynamic knee flexion and valgus angles

Figures 3 and 4 report the outcomes for dynamic knee flexion and valgus angles, respectively. All autograft groups demonstrated similar side-to-side differences in dynamic knee flexion angles ($p > 0.05$). For side-to-side differences in dynamic knee valgus angle, post hoc analyses revealed similar values among the QT and BPTB autograft groups, whereas the QT and HT autograft groups significantly differed between 0.23 and 0.30 s after IC of the SLHD ($p = 0.01$). The QT autograft group exhibited a smaller and the HT group

TABLE 1 Patient characteristics.

	QT	BPTB	HT	<i>p</i>
Number of patients (<i>n</i>)	10	10	10	
Age at follow-up (years)	31.5 ± 4.7	30.5 ± 6.1	31.0 ± 3.7	0.90
Sex (male/female)	9/1	7/3	8/2	0.54
Body mass index (kg/m ²)	25.2 ± 4.1	25.6 ± 2.7	26.2 ± 3.0	0.81
Injured side (dominant/non-dominant)	5/5	8/2	4/6	0.17
Time from injury to surgery (months)	4.7 ± 2.2	4.3 ± 1.8	5.2 ± 2.0	0.61
Medial meniscal injury at surgery (<i>n</i>)				0.29
No injury	9	7	9	
Yes, no treatment	1	1	0	
Yes, meniscectomy	0	2	0	
Yes, meniscus repair	0	0	0	
Lateral meniscal injury at surgery (<i>n</i>)				0.51
No injury	6	8	8	
Yes, no treatment	0	0	0	
Yes, meniscectomy	4	2	2	
Yes, meniscus repair	0	0	0	
Cartilage injury at surgery (<KL Grade II) (<i>n</i>)				0.33
No injury	10	8	7	
Medial	0	2	2	
Lateral	0	1	2	
Patellar	0	0	0	
Returned to sports after surgery (<i>n</i>)				0.63
Yes	9	8	8	
No, due to ACL injury	1	2	1	
No, due to other reason(s)	0	0	1	
Time from surgery to follow-up (years)	4.7 ± 1.7	4.4 ± 1.7	5.1 ± 2.0	0.66
Passive ATT at follow-up ^a (mm)				
Injured knee	6.3 ± 1.3	6.2 ± 1.6	5.5 ± 1.7	0.49
Non-injured knee	4.5 ± 1.5	3.8 ± 0.9	4.6 ± 2.0	0.52
STS difference (injured – non-injured)	1.8 ± 1.9	2.4 ± 1.3	0.9 ± 1.4	0.14

Note: Reported as mean ± SD, unless otherwise indicated.

Abbreviations: ACL, anterior cruciate ligament; ATT, anterior tibial translation; BPTB, bone-patellar tendon-bone; HT, hamstrings tendon; KL, Kellgren and Lawrence; LSI, limb symmetry index; QT, quadriceps tendon; SD, standard deviation; STS, side-to-side.

^aMeasured using the Lachmeter (Lachmeter®, Lachmeter Company) [13].

exhibited a larger knee valgus angle in the injured knee compared to the non-injured knee (maximal difference: QT = −3.90° vs. HT = +1.24°), mainly attributed to between-group differences in the non-injured knees (Figure 4). Furthermore, similar side-to-side differences in dynamic knee valgus angles were observed among the BPTB and HT autograft groups ($p > 0.05$).

DISCUSSION

The most important finding of this study was that at 4.5 years post-ACLR, side-to-side differences in dynamic tibiofemoral movements, expressed with ATTd and ITRd, during jump-landing tasks and walking, were similar in patients who underwent ACLR with a QT,

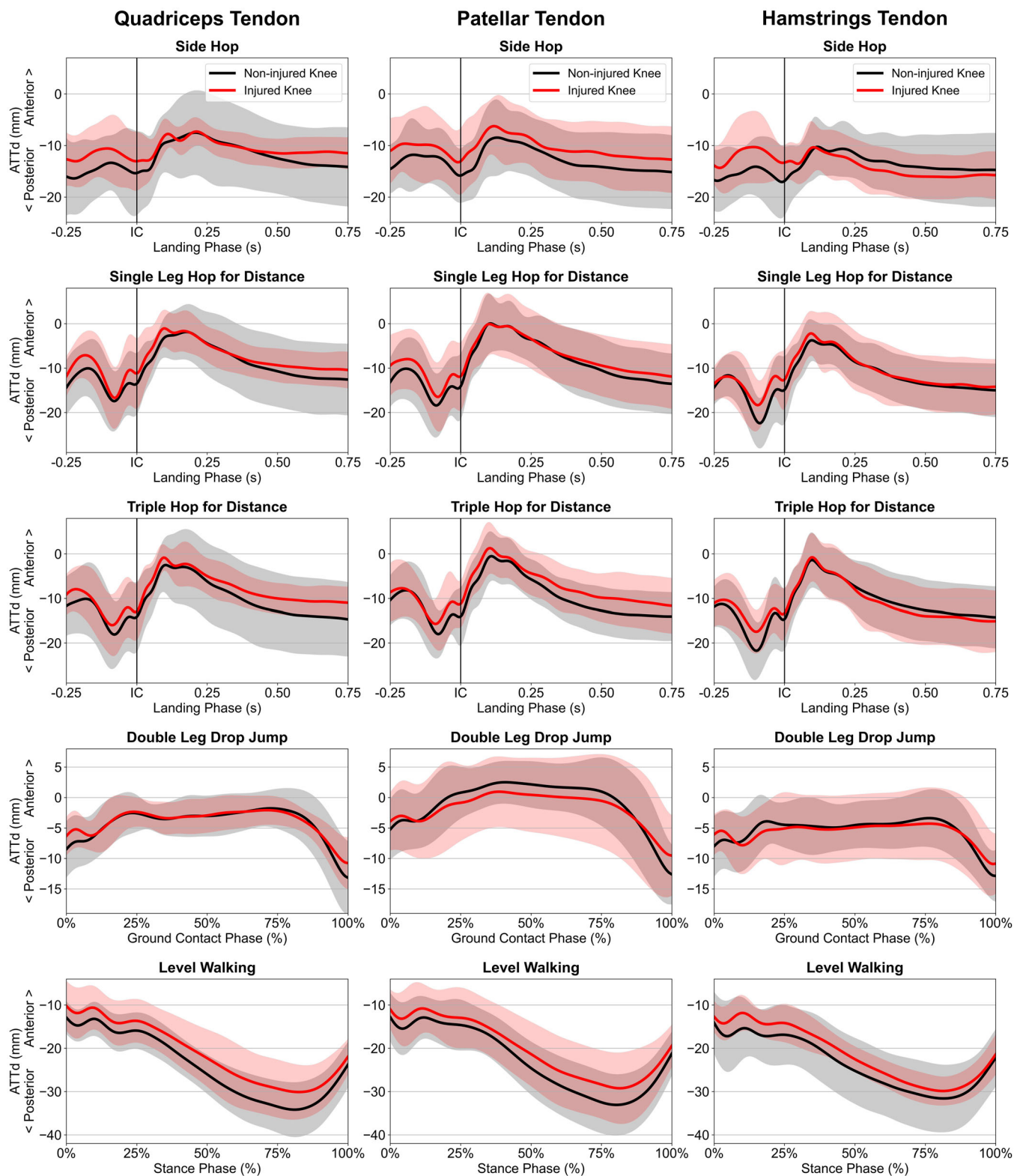


FIGURE 1 Dynamic anterior tibial translation (ATTd) in the injured (solid red line) and non-injured (solid black line) knee for each dynamic task and each autograft group, with standard deviations shown as shaded areas. An increasing value (less negative or more positive) indicates anterior tibial translation relative to the femur. The hop tasks are displayed from 0.25 s before initial contact (IC) to 0.75 s after IC, which is indicated with a vertical black line. The double-leg drop-jump is displayed as a percentage of the ground contact phase prior to take off. Level walking is displayed as a percentage of the stance phase. Only between-group comparisons of side-to-side differences were conducted, with no significant differences observed.

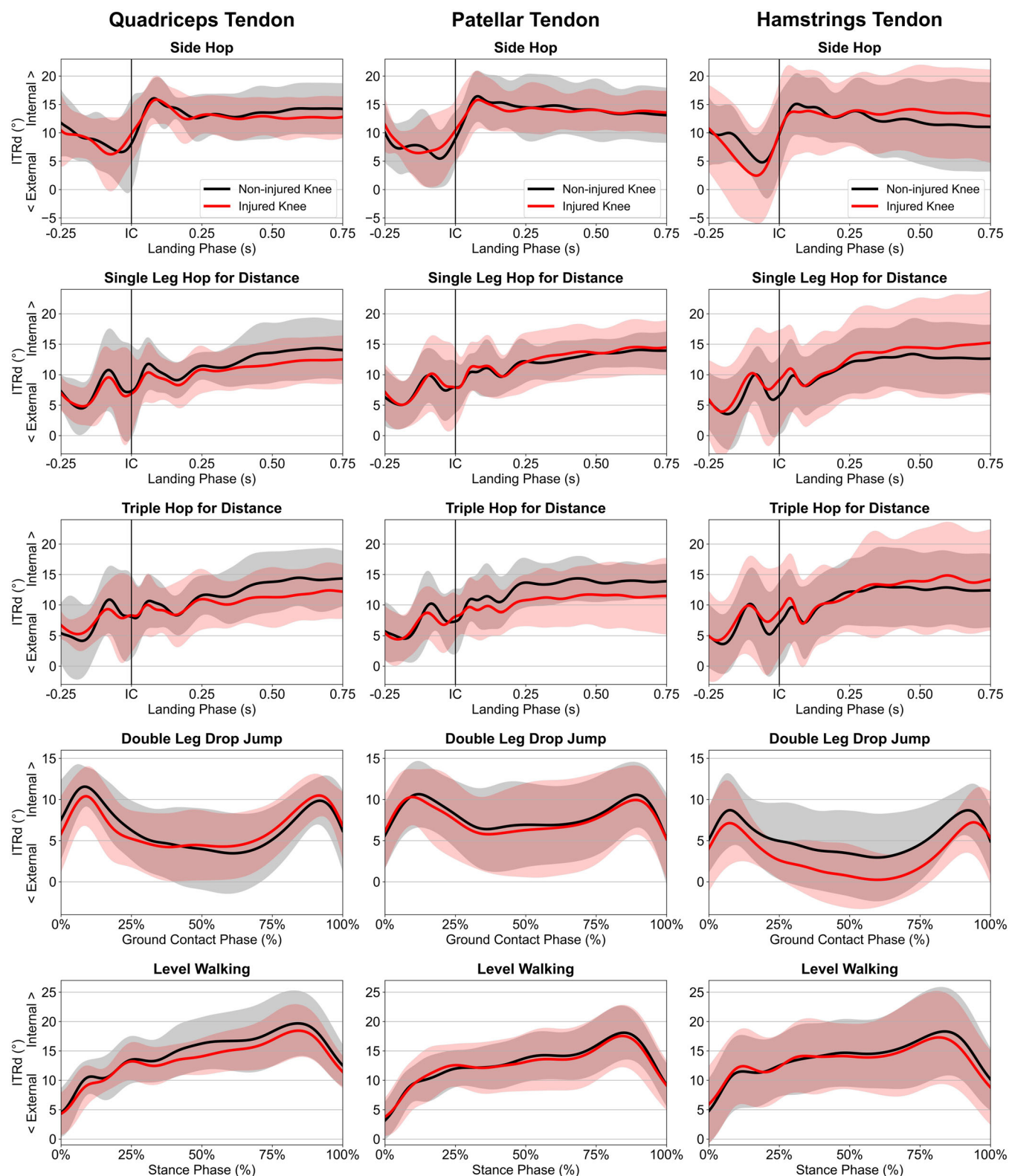


FIGURE 2 Dynamic internal tibial rotation (ITRd) in the injured (solid red line) and non-injured (solid black line) knee for each dynamic task and each autograft group, with standard deviations shown as shaded areas. An increasing value (less negative or more positive) indicates internal tibial rotation relative to the femur. The hop tasks are displayed from 0.25 s before initial contact (IC) to 0.75 s after IC, which is indicated with a vertical black line. The double-leg drop-jump is displayed as a percentage of the ground contact phase prior to take off. Level walking is displayed as a percentage of the stance phase. Only between-group comparisons of side-to-side differences were conducted, with no significant differences observed.

TABLE 2 Maximal isometric hamstrings and quadriceps strength.

	QT	BPTB	HT	<i>p</i>
Hamstrings strength				
Injured knee (N m)	98.1 ± 23.0	103.9 ± 30.0	89.2 ± 31.9	
Non-injured knee (N m)	98.9 ± 14.1	103.6 ± 20.0	114.0 ± 36.3	
LSI (injured/non-injured × 100%)	98.4 ± 12.7	99.3 ± 17.7	79.0 ± 18.3	*0.02^a
Quadriceps strength				
Injured knee (N m)	240.2 ± 48.0	222.2 ± 55.0	281.0 ± 90.6	
Non-injured knee (N m)	248.3 ± 55.3	231.4 ± 45.9	285.2 ± 85.6	
LSI (injured/non-injured × 100%)	98.8 ± 19.3	95.6 ± 9.7	98.9 ± 13.1	0.85
Hamstrings/quadriceps ratio				
Injured knee (%)	40.9 ± 5.6	47.2 ± 9.8	31.9 ± 5.7	
Non-injured knee (%)	41.1 ± 7.9	45.3 ± 6.1	40.3 ± 5.3	
LSI (injured/non-injured × 100%)	102.0 ± 17.9	104.6 ± 20.3	79.6 ± 13.5	*0.01^a

Note: Only the LSI was tested for statistical significance. Significant *p*-values are indicated in bold with an asterisk.

Abbreviations: BPTB, bone-patellar tendon-bone; HT, hamstrings tendon; LSI, limb symmetry index; QT, quadriceps tendon.

^aPost hoc test significant for QT versus HT versus and BPTB versus HT.

TABLE 3 Jumping distance and peak ground reaction forces.

	QT	BPTB	HT	<i>p</i>
Jumping distance (cm)				
Side hop				
Injured knee	98.2 ± 20.1	100.0 ± 20.6	113.7 ± 26.8	
Non-injured knee	98.6 ± 19.6	103.7 ± 22.0	112.7 ± 20.2	
LSI (injured/non-injured × 100%)	99.8 ± 8.0	96.8 ± 5.5	100.4 ± 9.9	0.59
Single leg hop for distance				
Injured knee	157.5 ± 15.6	151.1 ± 25.9	169.0 ± 28.8	
Non-injured knee	159.8 ± 20.0	155.1 ± 28.0	163.6 ± 29.3	
LSI (injured/non-injured × 100%)	99.0 ± 5.3	97.9 ± 6.5	103.5 ± 3.3	0.07
Triple hop for distance				
Injured knee	436.2 ± 70.9	418.2 ± 84.6	479.2 ± 92.6	
Non-injured knee	437.3 ± 68.5	429.7 ± 88.4	472.7 ± 91.7	
LSI (injured/non-injured × 100%)	100.0 ± 7.1	97.6 ± 6.1	101.6 ± 5.7	0.41
Peak vertical GRF (BW) ^a				
Side hop				
Injured knee	2.83 ± 0.50	2.85 ± 0.34	3.11 ± 0.62	
Non-injured knee	2.89 ± 0.57	2.76 ± 0.24	3.14 ± 0.76	
Difference (injured – non-injured)	−0.06 ± 0.37	0.09 ± 0.28	−0.03 ± 0.32	0.55
Single leg hop for distance				
Injured knee	3.19 ± 0.64	3.22 ± 0.52	4.08 ± 0.82	
Non-injured knee	3.05 ± 0.55	3.29 ± 0.35	3.66 ± 1.01	
Difference (injured – non-injured)	0.14 ± 0.36	−0.07 ± 0.41	0.42 ± 0.43	*0.04^b

(Continues)

TABLE 3 (Continued)

	QT	BPTB	HT	<i>p</i>
Triple hop for distance				
Injured knee	3.58 ± 0.79	3.33 ± 0.63	4.40 ± 0.64	
Non-injured knee	3.46 ± 0.62	3.61 ± 0.53	4.23 ± 1.04	
Difference (injured – non-injured)	0.13 ± 0.29	–0.28 ± 0.44	0.17 ± 0.56	0.06
Double-leg drop-jump				
Injured knee	1.95 ± 0.31	1.92 ± 0.35	2.04 ± 0.34	
Non-injured knee	2.11 ± 0.39	1.82 ± 0.42	2.09 ± 0.43	
Difference (injured – non-injured)	–0.16 ± 0.39	0.10 ± 0.32	–0.05 ± 0.39	0.30
Peak posterior GRF (BW) ^a				
Side hop				
Injured knee	0.58 ± 0.17	0.58 ± 0.19	0.65 ± 0.27	
Non-injured knee	0.59 ± 0.13	0.52 ± 0.14	0.68 ± 0.23	
Difference (injured – non-injured)	–0.01 ± 0.15	0.05 ± 0.09	–0.04 ± 0.14	0.29
Single leg hop for distance				
Injured knee	1.07 ± 0.24	1.01 ± 0.27	1.26 ± 0.31	
Non-injured knee	1.03 ± 0.30	1.04 ± 0.28	1.11 ± 0.32	
Difference (injured – non-injured)	0.04 ± 0.25	–0.03 ± 0.13	0.15 ± 0.22	0.18
Triple hop for distance				
Injured knee	1.31 ± 0.38	1.16 ± 0.37	1.46 ± 0.26	
Non-injured knee	1.31 ± 0.31	1.34 ± 0.38	1.42 ± 0.27	
Difference (injured – non-injured)	–0.003 ± 0.28	–0.18 ± 0.19	0.04 ± 0.20	0.08
Double-leg drop-jump				
Injured knee	0.45 ± 0.12	0.44 ± 0.11	0.52 ± 0.11	
Non-injured knee	0.43 ± 0.12	0.48 ± 0.17	0.49 ± 0.10	
Difference (injured – non-injured)	0.02 ± 0.08	–0.05 ± 0.12	0.03 ± 0.06	0.12

Note: Only the LSI and the difference (injured – non-injured) were tested for statistical significance. Significant *p*-values are indicated in bold with an asterisk. Abbreviations: BPTB, bone-patellar tendon-bone; BW, body weight; GRF, ground reaction force; HT, hamstrings tendon; QT, quadriceps tendon.

^aGRFs were normalized to BW.

^bPost hoc test significant for BPTB versus HT.

BPTB and HT autograft. Patients with a QT autograft demonstrated similar isometric hamstrings strength LSI and H/Q ratio LSI compared to patients with a BPTB autograft, whereas both these autograft groups demonstrated greater hamstrings strength LSI and H/Q ratio LSI compared to patients with an HT autograft. Isometric quadriceps strength LSI was similar among autografts.

Due to differences in biomechanical properties, graft fixation and H/Q strength ratio among various graft types for ACLR [16, 17, 23, 46], there has been an ongoing debate over the past decades regarding which graft provides the greatest post-operative knee stability [38, 39]. Previous studies have reported

comparable passive ATT values for QT, BPTB and HT autografts [11, 25, 32], and comparable short-term dynamic tibiofemoral movements for the BPTB and HT autografts during walking, stair descent and jump-landing tasks [3, 21, 52, 53]. The findings of the present study confirmed the similarity in dynamic tibiofemoral movements among BPTB and HT autografts and demonstrated that this similarity persists at mid-term follow-up. Moreover, the present study demonstrated comparable dynamic knee stability following ACLR with a QT autograft compared to the BPTB and HT autografts. The observed effect sizes were small or on the small-to-medium threshold, suggesting that differences in ATTD and ITRd among

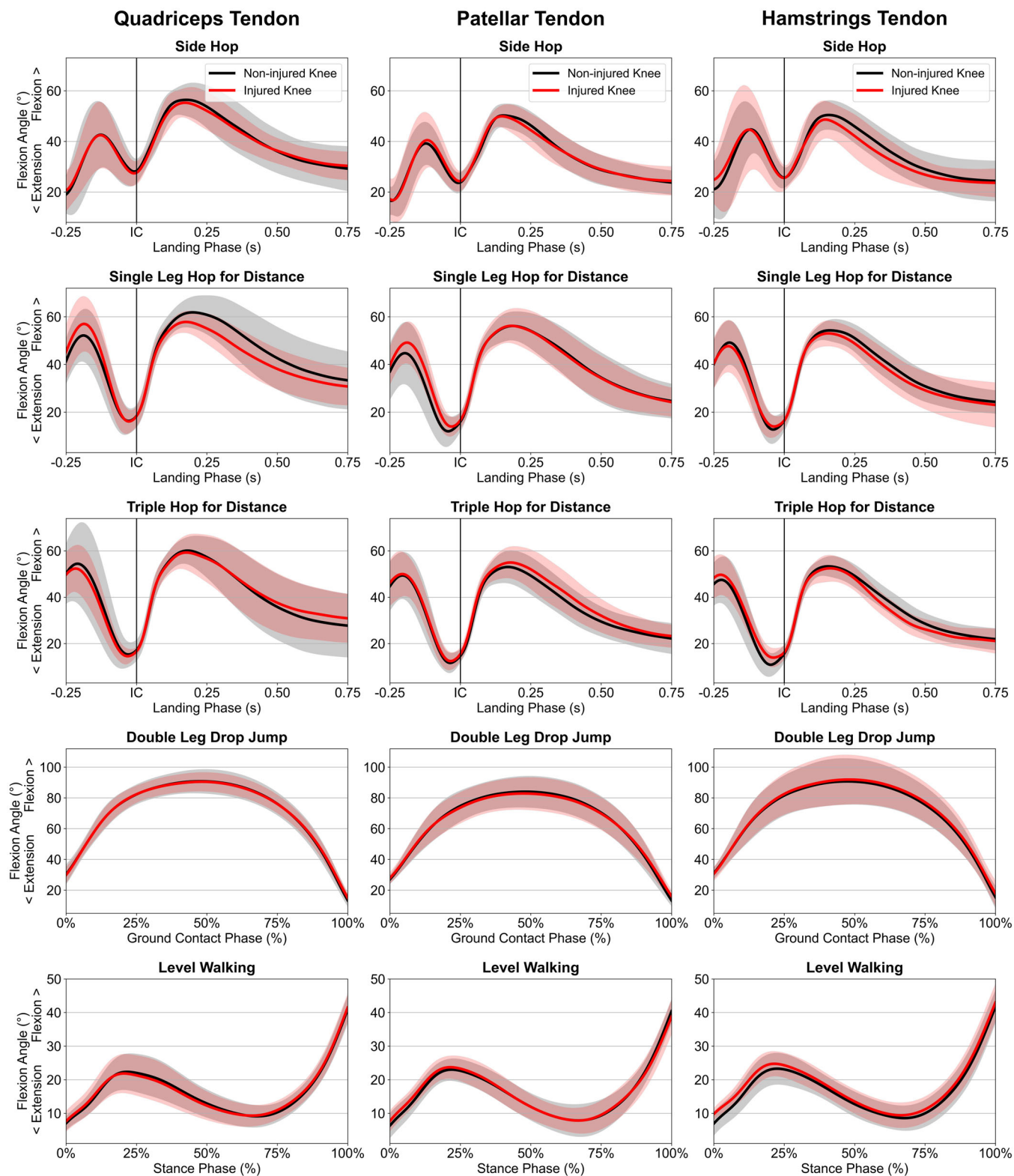


FIGURE 3 Dynamic knee flexion angles in the injured (solid red line) and non-injured (solid black line) knee for each dynamic task and each autograft group, with standard deviations shown as shaded areas. An increasing value indicates increased knee flexion. The hop tasks are displayed from 0.25 s before initial contact (IC) to 0.75 s after IC, which is indicated with a vertical black line. The double-leg drop-jump is displayed as a percentage of the ground contact phase prior to take off. Level walking is displayed as a percentage of the stance phase. Only between-group comparisons of side-to-side differences were conducted, with no significant differences observed.

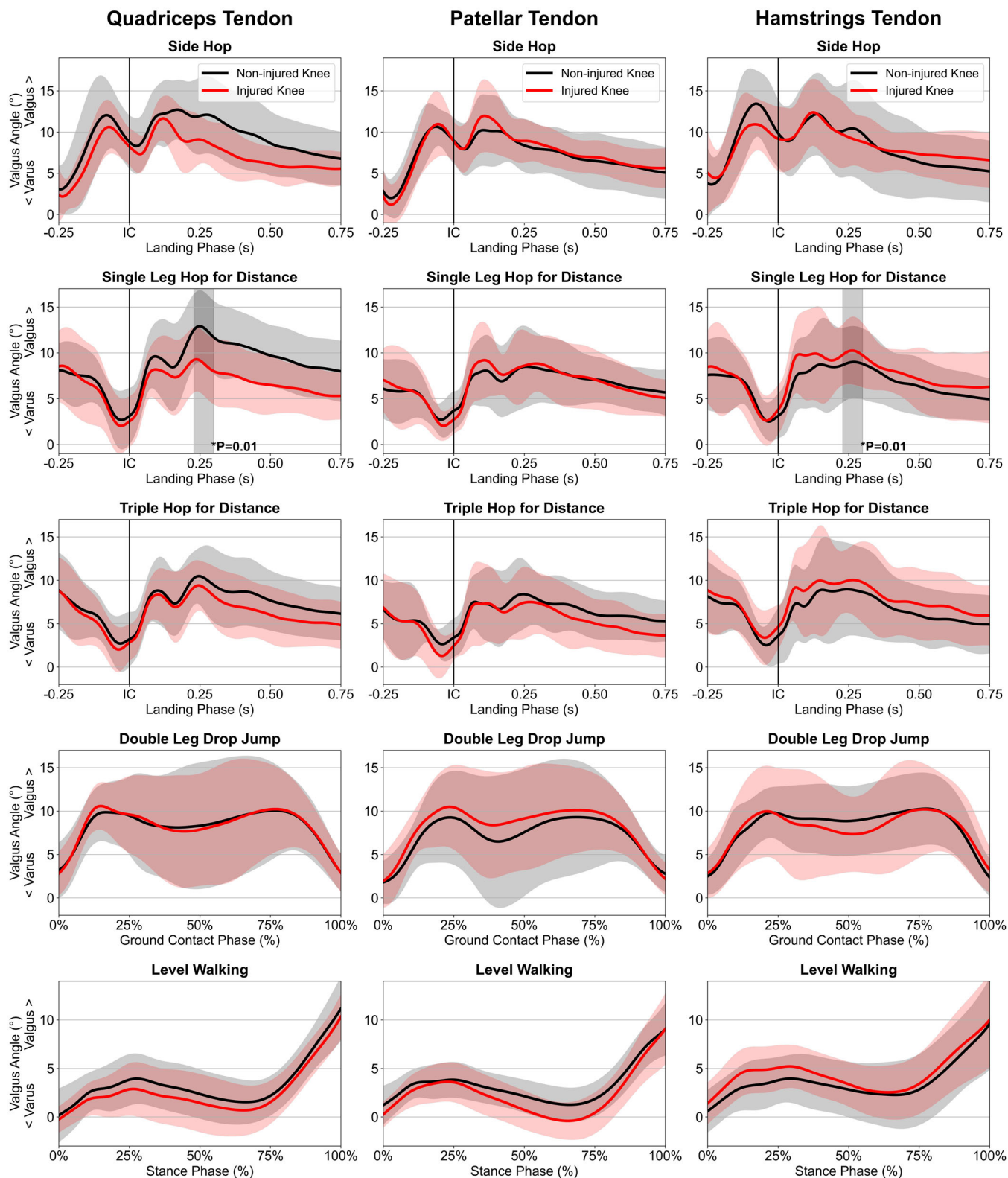


FIGURE 4 Dynamic knee valgus angles in the injured (solid red line) and non-injured (solid black line) knee for each dynamic task and each autograft group, with standard deviations shown as shaded areas. An increasing value indicates increased knee valgus. The hop tasks are displayed from 0.25 s before initial contact (IC) to 0.75 s after IC, which is indicated with a vertical black line. The double-leg drop-jump is displayed as a percentage of the ground contact phase prior to take off. Level walking is displayed as a percentage of the stance phase. The quadriceps and hamstrings tendon groups differed significantly in side-to-side difference in knee valgus angle during the landing of the single leg hop for distance ($p = 0.01$), indicated by the grey-shaded area.

autograft types were unlikely to be clinically meaningful. These results complement previous evidence supporting the QT autograft as a viable option for ACLR, demonstrating reduced anterior donor-site pain compared to the BPTB autograft, better functional outcome (Lysholm) scores compared to the HT autograft and similar graft failure rates compared to the BPTB and HT autografts [32]. Furthermore, since alterations in dynamic tibiofemoral movements are associated with post-traumatic osteoarthritis [51], the observed similarity in ATTD and ITRd among all three autografts may help explain the observed comparable incidence of post-traumatic osteoarthritis following ACLR with BPTB and HT autografts [50], and could imply that a similar incidence might also be expected in patients with QT autografts, which has not yet been investigated.

In contrast to similar between-group ATTD and ITRd values, several between-group differences in lower extremity strength values were evident in the present study. Patients with an HT autograft demonstrated lower hamstrings strength LSI and H/Q ratio LSI compared to patients with a QT and BPTB autograft, while all autograft groups demonstrated similar quadriceps strength LSI. This is consistent with prior research, reporting persistent hamstring strength deficits in patients with an HT autograft at 2-year follow-up, whereas quadriceps strength deficits in patients with QT and BPTB autografts were reported to be resolved after 8 months post-ACLR [6, 23]. The results of the present study suggest that hamstring weakness due to graft harvesting in patients with an HT autograft persists even 5 years after surgery, while patients with a QT and BPTB recover their loss in quadriceps strength. Reduced H/Q strength ratio has been identified as a key risk factor for ACLR graft failure [27], highlighting the need to target hamstrings weakness in patients with an HT autograft in injury prevention and return-to-sports protocols. Meanwhile, it is important to recognize that, despite restored quadriceps strength, QT and BPTB autograft harvest can still compromise neuromuscular function by disrupting sensory feedback, as evidenced by persistent proprioception deficits and prolonged quadriceps activation latency following harvest [15, 22]. Although these impairments did not appear to affect ATTD and ITRd in the present study, impaired proprioception following graft harvest has been suggested as a potential contributor to ACL re-injury risk and the development of osteoarthritis [22], emphasizing the importance of graft-specific rehabilitation strategies.

Several studies have already emphasized the need to address hamstring weakness after ACLR with an HT autograft. The hamstrings appear to unload the ACL by generating posterior tibial shear force, and hamstrings-quadriceps co-contraction can have a protective effect by reducing ATT and ITR [30]. Additionally, a cadaver

study reported that unloading of the medial hamstrings, which simulates a gracilis and semitendinosus tendon harvest, resulted in an anteriorly translated and externally rotated tibia [47]. However, in the present study, hamstring weakness in the HT autograft group did not appear to influence ATTD and ITRd. A potential explanation might be that the generated forces on the knee during the dynamic tasks were insufficient to elicit measurable differences. The needed hamstring strength required to counteract these forces may lie within their physiologically achievable range. However, this suggests that patients with an HT autograft may need to recruit their hamstrings closer to maximal activation to maintain similar knee kinematics. Such elevated hamstring demands are reported to result in greater muscle fatigue [10], which in turn alters both the latency and magnitude of hamstrings reflex responses [1]. This could eventually contribute to increased ATTD and ITRd over time as muscle fatigue progresses [1], and may provide an explanation for the previously reported higher incidence of graft failure in patients with a HT autograft [39]. Therefore, hamstring weakness in patients with an HT autograft might be an important factor to consider during graft selection and should be addressed in injury prevention and return-to-sports protocols, especially for patients engaged in sports that demand high levels of knee flexion strength [37]. As the present study did not assess muscle activation or include a fatigue condition, future research is necessary to confirm this hypothesis.

The type of autograft did not have a substantial effect on dynamic knee flexion and valgus angle. While the present study did find a significant difference in side-to-side dynamic knee valgus angle during the SLHD, the observed difference occurred during the late landing phase, coinciding with extension of the knee after the largest forces have been controlled. Additionally, the lower side-to-side difference in dynamic knee valgus angle in the QT autograft group was primarily due to increased knee valgus in the non-injured leg rather than decreased knee valgus in the injured leg, suggesting that this asymmetry was unrelated to the autograft type. These findings are mostly consistent with existing literature, reporting no difference in dynamic knee flexion and valgus angles during the DLDJ among QT, BPTB and HT autografts [19, 33, 48]. In contrast, the HT autograft group in the present study demonstrated a larger side-to-side difference in peak vertical GRF during the SLHD compared to the BPTB autograft group. A potential explanation might be the observed hamstrings weakness in the HT autograft group, which is reported to be associated with higher vertical GRF during landing tasks [9]. Increased vertical GRF is related to a biomechanical profile associated with greater ACL loading and may therefore reflect an increased ACL injury risk [29, 54].

The main strength of this study is the inclusion of patients who were randomly allocated to the three most

commonly used autograft types for ACLR within a single study, strengthening the level of evidence. However, this study also includes several limitations. Firstly, the study was limited by a relatively small sample size within each group, which may have masked subtle differences between autografts and increased the risk of a Type II error, despite an a priori sample-size calculation. Nevertheless, the reported effect sizes were mostly small, or at the small-to-medium boundary, suggesting that the observed between-group differences are likely minor and may not be clinically meaningful. In addition, due to selective re-recruitment from a previously conducted RCT, potential selection bias may exist. The included patient cohort was assessed at mid-term follow-up, had not experienced graft failure or contralateral ACL injury, had mostly returned to sports and demonstrated high jumping distance LSI, indicating they were well-rehabilitated. Therefore, generalization of the present study's findings is limited to the specific included patient population, and differences in ATTD and ITRd between autograft types may have existed in patients who sustained graft failure prior to the measurement. Furthermore, although marker-based motion analysis is often considered the practical standard for 3D kinematic assessment, it is more susceptible to soft tissue artefacts compared to fluoroscopy (the gold standard). Applying several techniques to reduce these artifacts—including a large number of markers distributed over the femur and tibia in combination with a rigid body marker configuration [12, 43], and analyzing side-to-side differences rather than absolute values—enabled dynamic tibiofemoral movements to be captured during demanding tasks with full range of motion and sufficient accuracy. Nevertheless, the interpretation of ATTD and ITRd findings warrants caution. Additionally, although the assessment of hamstrings and quadriceps strength alongside dynamic tibiofemoral movements is a strong point of this study, muscle strength was assessed isometrically, which may not fully reflect strength during dynamic tasks, as isokinetic assessment would. However, previous studies have reported a strong correlation between isometric and isokinetic strength assessments [4, 18]. Future research should examine dynamic tibiofemoral movements in larger cohorts and in patients with poorer ACLR outcomes, such as those who experienced graft failure, potentially with fluoroscopy, alongside isokinetic hamstrings and quadriceps strength.

CONCLUSIONS

Patients who underwent ACLR with a QT autograft demonstrated similar side-to-side differences in ATTD and ITRd during jump-landing tasks and level walking compared to patients with BPTB and HT autografts 5 years post-ACLR. Furthermore, patients with QT and BPTB autografts showed no persistent

graft-harvest-related strength deficits, whereas patients with an HT autograft exhibited persistent hamstrings weakness 5 years post-ACLR. Routine graft selection should consider hamstrings strength deficits in patients with an HT autograft, while ATTD and ITRd are not autograft-dependent. These findings support QT autografts as a viable option for ACLR.

AUTHOR CONTRIBUTIONS

Tom Vendrig: Conceptualization; methodology; formal analysis; investigation; writing—original draft preparation. **Michèle N. J. Keizer:** Conceptualization; methodology; writing—review and editing. **Reinoud W. Brouwer:** Conceptualization; methodology; writing—review and editing. **Han Houdijk:** Conceptualization; methodology; writing—review and editing. **Roy A. G. Hoogeslag:** Conceptualization; methodology; writing—review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author.

ETHICS STATEMENT

Approved by the institutional review board of the University Medical Center Groningen (UMCG) (2022.349). Prior to enrolment, all participants were informed about the study protocol and signed informed consent.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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